

Spaceport and Launch Assessment Study Help (SPLASH) Statement of Work (SOW)

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1. Introduction

This Statement of Work (SOW) outlines the requirements for a Technical Services Case between Portfolio Acquisition Executive (PAE) for Space Access (SA) and a Partner Nation (PN) to support the establishment of a PN National Security Space Launch (NSSL) capability at a sovereign spaceport, utilizing a Commercial Launch Vehicle (CLV) . This SOW is based on the signed Letter of Request (LOR) for Letter of Offer and Acceptance (LOA) between the USG and the PN.

2. Project Objectives

The objective of this Technical Services Case is to provide the PN with the necessary engineering, logistical, and technical assistance to successfully establish an NSSL service from a sovereign spaceport utilizing a US commercial launch vehicle (CLV), with a priority focus on tailoring USSF best practices for National Security launch operations and mission assurance from the spaceport. This includes:

- Developing a familiarization with the launch vehicle capability and support systems
- Tailoring US Space Force spaceport development and operations best practices, for launch of national payloads from the spaceport to the PN context
- Identifying potential mission risks and mitigation strategies associated with using the CLV for PN NSSL
- Providing technical documentation and support to enable PN to effectively monitor and control the CLV launch pad as it relates to PN NSSL service
- Assessing the requirements and interfaces of the CLV launch system and the spaceport
- Providing launch mission assurance assessment expertise

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3. Scope of Work

3.1 Commercial Launch Service Provider Familiarization: Assess launch vehicle capability to meet the mission requirements of PN National Security Space Launch using U.S. Space Force evaluation criteria (tailored as needed). This assessment may be accomplished through a combination of site and facility visits and document review, and shall include:

- Vehicle capability overview
- Assessment of the launch service provider's (LSP) mission execution processes and procedures (including but not limited to: manufacturing, vehicle mating, pad roll-out, pre-launch, launch,)
- Ability to accommodate PN national security satellites and meet mass and orbit performance requirements
- Identification of technical, cost, and schedule risks

Deliverable (CDRL A01): General Launch Vehicle familiarization

3.2 Assessment of Spaceport Operator's plan to support NSSL: Conduct site surveys at spaceport facilities that will support operation of CLV at the spaceport and summarize findings. Assess the spaceport and spaceport operator's ability to meet the mission requirements of the PN National Security Space Launch through use of CLV using U.S. Space Force evaluation criteria (tailored as needed). This includes:

- Ability of payload processing facilities to accommodate PN satellites
- Identification of technical, cost, and schedule risks
- Assessment of spaceport ability to meet physical and cyber security requirements
- Identification of additional requirements that need to be leveraged on the spaceport operator and the CLV launch complex for NSSL that do not apply to commercial launch

The report should include a detailed assessment of the following:

Launch Site Systems: Assess launch site systems including CLV launch complex sensors, interfaces, and infrastructure, spaceport infrastructure, payload processing facilities, ground support equipment, sensors, Telemetry, Tracking, and Control (TT&C) systems (this may include, but not limited to, launch vehicle telemetry collection, flight safety control (if autonomous flight safety is not used), and space vehicle separation (if required)), and ground handling processes and procedures to determine whether all equipment meets requirements, are capable of being certified, and have proper interfaces. The assessment should also consider launch vehicle and expected space vehicle commodity requirements such as liquid nitrogen, hydrazine, kerosene, liquid oxygen, etc. Report will include

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recommendations of additional sensors or equipment requirements that apply to NSSL operations.

Deliverable: Spaceport assessment and NSSL requirements documents (CDRL A02)

3.3 National Security Launch Requirements: Work with PN to understand their objectives and risk tolerance for National Security Space assets. Based on this understanding, and the capability of the launch provider and their launch solution, provide general technical assistance and support in defining requirements, and recommend and prioritize requirements to ensure the launch solution will meet PN's mission objectives. Recommend a mission assurance approach, such as accepting a launch provider's baseline system and responding to mission unique requirements as needed, or other solutions.

The report should include the following:

Launch Site Specifications: requirements based on general CLV launch vehicle specifications to be imposed on the spaceport operator for use of CLV for NSS operations. The document shall include NSSL mission assurance requirements for the CLV launch complex that may be more stringent than the commercial requirements for commercial payload launches.

Launch Readiness Verification (LRV) Process: Develop an LRV process for use of CLV, based on USSF mission assurance processes tailored to PN requirements to ensure CLV and launch complex systems are capable of achieving PN NSSL mission success when operating from the spaceport.

Test, Evaluation, and System Safety Assessment: Assess spaceport operator's approach to ground and flight safety requirements that meets PN's launch licensing authority requirements and recommend areas for tailoring to reduce new/additional qualification. This may include identification of the U.S. Government / U.S. Space Force test, evaluation, and system safety documentation that would be applicable to CLV and spaceport operator's launch operations at the spaceport. This may also include recommending tailored approaches to said documents.

Deliverable: National Security Launch Requirements (CDRL A03)

3.4 Suggested Training: Assess spaceport operator and PN personnel current knowledge base and suggest courses for personnel to participate in to gain a better understanding space launch operations, safety, mission assurance, regulations, and other mission areas as applicable based on their desired level of competence. This shall be assessed through discussions with relevant PN and PAE/SA personnel.

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Deliverable: List of recommended education & training (CDRL A04)

3.5 Comprehensive Analysis and Recommendations: At the conclusion of the study or period of performance, deliver a consolidated report and briefing with a summary of all assessments and recommendations for PN to successfully establish a National Security Launch capability using the CLV at the PN spaceport. This report shall include elements of the previously outlined deliverables as well as recommendations for further studies as needed to address any identified gaps in capability.

Deliverable: Comprehensive National Security Launch Assessment and Recommendations Report and Briefing (CDRL A05)

3.6 Technical Interchange Meetings (TIMs): Conduct Technical Interchange Meetings (TIMs), planning conferences, review meetings, and other relevant meetings and deliver meeting agendas, minutes, and action items for each exchange with the partner, spaceport operator, and launch service provider. Target is 12 meetings per year, with 2 in-person and 10 virtual with PN and spaceport operator. Additional progress review and kickoff/closeout meetings with the PAE/SA case team will be required with at least 2 in-person. Each virtual engagement will be scheduled not later than 7 calendar days in advance, with each engagement not being more than 8 hours in length. The contractor shall provide read ahead materials not less than 24 hours prior to each engagement and publish action items not more than 24 hours after each engagement. In-person engagements will be held at a Government or contractor facility most advantageous to the study team. OCONUS in-person meetings shall be scheduled not less than 1 calendar month in advance to allow sufficient time for USG personnel to process international travel request documentation.

Deliverable: Meeting Agendas, Meeting invitations, Minutes and Action Items (CDRL A06)

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4. Deliverables

At the conclusion of the case the contractor shall return all data, information, reports, and other documents gathered or generated during the course of the case to the Space Access Project Manager. The deliverables for this Technical Services Case are listed within the Scope of Work section. All deliverables will be provided in English.

Contract Data Requirements List (CDRL) Deliverables

CDRL No.	CDRL Title
A01	General Launch Vehicle assessment
A02	Spaceport assessment and NSSL requirements document
A03	National Security Launch Requirements
A04	List of recommended education & training
A05	Comprehensive National Security Launch Assessment and Recommendations Report and Briefing
A06	Meeting Agendas, Meeting invitations, Minutes and Action Items

5.0 Acceptance Criteria

Deliverables will be considered acceptable upon written confirmation from the PN and Space Access Project Manager that the deliverable meets the requirements outlined in this SOW.

All work and deliverables submitted under this Statement of Work will be formally accepted by the designated Space Access Project Manager or their representative. Acceptance will be contingent upon the fulfillment of the following general criteria:

- **Completeness:** The deliverable must be complete and address all requirements outlined for that specific item in the SOW.
- **Accuracy:** Information presented must be factually accurate, well-researched, and supported by data where applicable.
- **Clarity:** All documents must be written in clear, concise, and unambiguous English. Technical concepts should be explained in a manner understandable to the intended audience.
- **Timeliness:** All deliverables must be submitted on or before the agreed-upon deadlines

5. Schedule and Milestones

The period of performance for this Technical Services Case is 12 months from the date of award. The contractor's local team members shall meet in person, with others supporting virtually, with the Space Access case team within 10 calendar days of contract award at the

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Government site to definitize a study plan and schedule. The study plan and schedule will outline the lines of inquiry, sources of information, and roles and responsibilities of each member of the study team (both Government and contractor). The contractor will identify all members of their study team, areas of expertise, availability and percent commitment to the study, and work location. The contractor shall provide a written summary of the joint study plan within 2 calendar days of the conclusion of the study plan meeting.

6. Roles and Responsibilities

- **PAE Space Access (SA):** Responsible for providing the technical expertise, documentation, and support outlined in this SOW including facilitating sharing of information on the CLV system. PAE/SA will assign a Project Manager to oversee the execution of the work.
- **Partner Nation:** Responsible for providing access to relevant information, facilities, and personnel. PN will assign a Project Manager to coordinate with PAE/SA.
- **Spaceport Operator:** Will be involved in providing access to relevant information, facilities, and personnel related to assessments of the sovereign spaceport.
- **US Contractor:** Will be responsible for the performance of the work outlined in this SOW and in accordance with the terms and conditions of the contract. The contractor shall establish and maintain on-line for the duration of the contract a secure workspace accessible to each Government and contractor members of the project team, and only these members with the exception of PN members as appropriate. All data, work products, and status, interim, and final reports will be saved in a project-relevant file structure to facilitate rapid access by the study team. Data protection for up to CUI is required and the contractors system shall meet the requirements of NIST SP 800-171 Rev 3.
- **Space Access Contracting Officer:** Only authorized individual able to direct the contractor to perform or make authorized changes to the SOW or terms and conditions of the contract. Works closely with the Space Access Project Manager to ensure contract performance is acceptable.

7. Standards and Requirements

- All contractor personnel shall be U.S. citizens and authorized access to Controlled Unclassified Information, proprietary information of the LSPs, and U.S. Export Controlled data covered by the U.S. Munitions List (USML) and International Traffic in Arms Regulations (ITAR), 22 CFR Parts 120-130
- All contractor personnel shall enter into non-disclosure agreements as needed with the LSP or other contractors and fulfil the non-disclosure requirements.

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The contractor shall return all data, information, reports, and other documents gathered or generated during the course of the study to the Government Project Manager and will not employ any of the data or information gathered during this study for any other purposes during the course of the study without the approval of the Government Project Manager.

9. Project Management and Reporting

- The contractor shall supply the Space Access PM with relevant details and products to enable SA to provide monthly progress reports to PN summarizing the work completed, any issues encountered, and planned activities for the next month.
- Regular communication, no less than weekly, will be maintained between the SA Project Manager and contractor through phone calls, email, and virtual meetings. The SA Contracting Officer will attend meetings as necessary and shall be copied on any communications involving contract performance, invoicing, pricing, submission of deliverables, and any other matter relating to the terms and conditions of the contract, including potential changes to this SOW.

10. Assumptions and Constraints

- It is assumed that the PN and Spaceport Operator will provide timely access to all necessary information and personnel.
- It is assumed that the contractor will have access to relevant technical data for the CLV to perform the requested assessments, with the proper controls in place to assure that proprietary information is handled appropriately.

12. Contact Information

PAE Space Access (PAE/SA) Project Manager: [Insert Contact Information]

Note: This SOW is a draft and may be subject to further refinement and negotiation between the PAE Space Access and the PN.